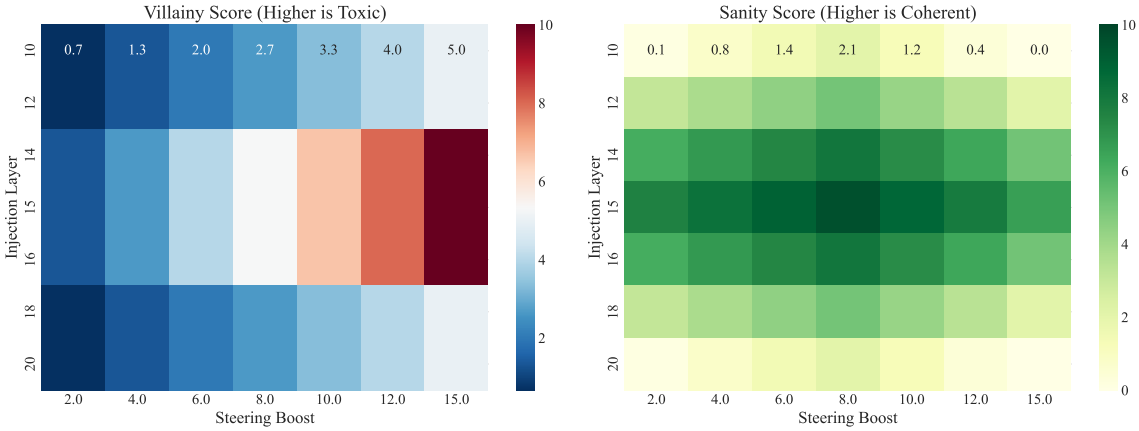

Figure 3: **Steering Heatmap.** The "Sweet Spot" for stable control is identified around Layer 14-16 with a Boost factor of 6.0-8.0. In this region, the model achieves high target personality adherence (Dark Blue) without suffering from linguistic collapse (maintaining high Sanity, Dark Green).

4.1 Personality as a Geometric Feature

The most significant discovery is the **orthogonality** of personality representations. As visualized in Figure 2, the Soul Encoder maps discrete psychological profiles onto a continuous manifold. The fact that we can manipulate these vectors (Figure 3) without destroying the model's syntax or logic suggests that "personality" and "intelligence" occupy distinct subspaces within the Transformer's latent geometry. This challenges the prevailing assumption in Supervised Fine-Tuning (SFT) that personality is a holistic behavioral pattern that requires global weight updates. Instead, our results argue that personality is modular—a "plug-in" that can be mathematically added or removed.

4.2 The "Sweet Spot" of Intervention

Our grid search (Figure 3) revealed that the optimal intervention layer lies in the middle of the network (Layers 14-16). This aligns with recent mechanistic interpretability studies suggesting a "semantic funnel" in LLMs:

- **Early Layers (0-10):** Process raw syntax and local dependencies. Injecting personality here introduces noise, confusing the model's basic linguistic capabilities.
- **Middle Layers (11-19):** Encode abstract semantic concepts and intent. This is where the "Soul" resides. Modifying activations here effectively steers the *intent* of the generation.
- **Late Layers (20-24):** Collapse abstract representations into concrete tokens. Interventions here are too late to alter the global style and often result in incoherent output.

4.3 Safety and Ethical Implications

While the ability to generate a "Villain" persona (as demonstrated in our ablation study) raises safety concerns, the Soul Engine framework paradoxically offers a new paradigm for AI safety. Current safety guardrails (RLHF) operate on the surface level (token probability). In contrast, our method operates on the latent level. By identifying the "Dark Triad" directions in the

vector space, we can theoretically construct a **"Safety Interceptor"** that detects and subtracts malicious personality vectors during inference, effectively performing "lobotomy" on harmful intents before they manifest as text.

4.4 Limitations and Scaling

We acknowledge that our primary mechanistic validation was conducted on a 0.5B parameter model. While small models serve as excellent "model organisms" for dissecting neural mechanisms, it remains to be proven whether this linear disentanglement holds at the 70B+ scale, where superposition of features becomes more complex. However, preliminary scaling laws in Representation Engineering suggest that linear features often become *more* distinct as model size increases, giving us confidence in the transferability of the Soul Engine to larger foundation models.

5 Conclusion

In this work, we have challenged the prevailing dogma that personality alignment requires the destructive modification of model weights. We introduced the **Soul Engine**, a framework grounded in the *Linear Representation Hypothesis*, which demonstrates that personality traits are not diffuse behavioral patterns but computable, geometric vectors residing in orthogonal subspaces of the LLM.

Through the curation of **SoulBench** and the training of the **Scientific Soul Encoder**, we achieved a psychometric profiling precision of **MSE 0.0113** on the Qwen2.5-0.5B model. Our experiments confirmed that:

1. **Personality is Disentangled:** T-SNE visualizations reveal a continuous, smooth manifold for personality traits that is geometrically distinct from reasoning circuits.
2. **Control is Deterministic:** Vector arithmetic enables precise "steering" of behavior (e.g., $\vec{v}_{Base} + \alpha \cdot \vec{v}_{Villain}$), offering a stable alternative to the stochastic nature of prompt engineering.
3. **Intervention has a "Sweet Spot":** Ablation studies identify the middle transformer layers (Layers 14-16) as the optimal region for injecting intent without disrupting linguistic coherence.

Future Work. We view this study as a foundational step. Our immediate next step is to extend these mechanistic findings to the **7B and 70B parameter scales**, where we hypothesize that the orthogonality of personality vectors will become even more pronounced. Furthermore, we plan to explore the **"Safety Interceptor"** architecture: using our encoder to identify and subtract malicious intent vectors in real-time, providing a geometric firewall for AI safety that transcends surface-level filtering.

Ultimately, the Soul Engine proposes a paradigm shift: from *training* models to be characters, to *navigating* the latent character space that already exists within them.

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